

Superheterodyne Receiver Link-Budget Analysis Using Keysight SystemVue

Project Objective:

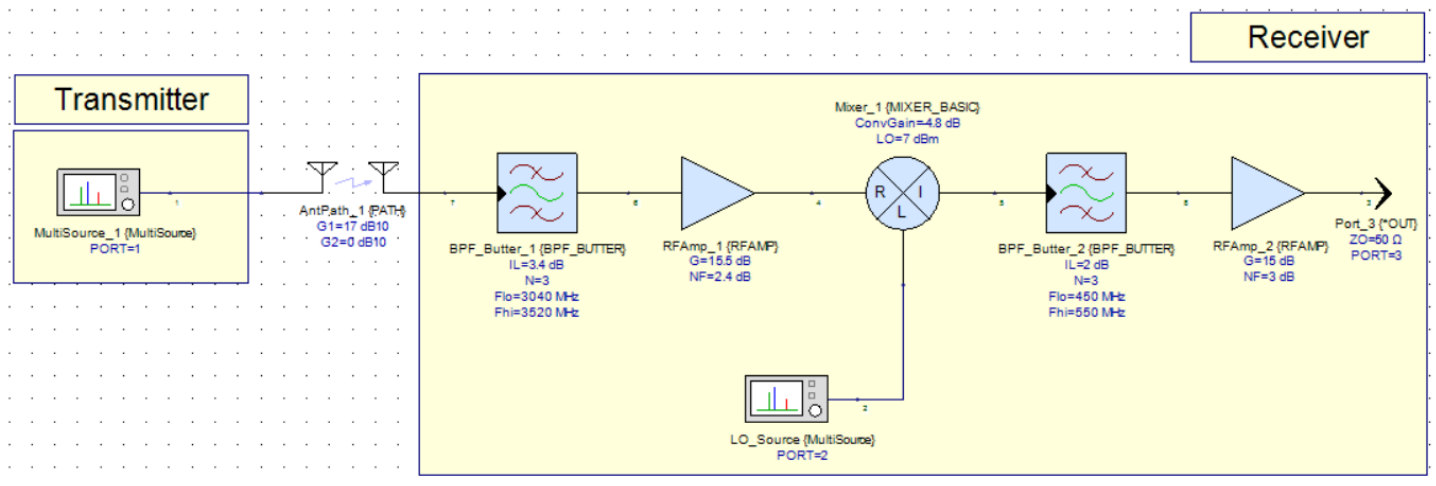
The objective of this project is to design and analyze a representative **superheterodyne receiver** using **Keysight SystemVue**.

The analysis considers a 5G n78 downlink operating at **3.5 GHz** and evaluates the signal power as it propagates through:

- Free-space propagation
- RF bandpass filter
- Low-noise amplifier (LNA)
- Mixer
- IF bandpass filter
- IF amplifier

The project also evaluates the receiver noise performance, noise floor, sensitivity, SNR, and link margin.

System Architecture:



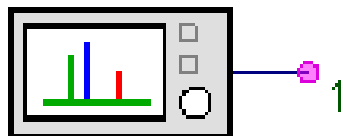
System Specifications:

Parameter	Value
Application	5G NR n78
RF frequency	3.5 GHz
Conceptual channel bandwidth	100 MHz
TX power	46 dBm
TX antenna gain	17 dBi
RX antenna gain	0 dBi
Distance	1 km
Reference impedance	50 Ω
LO frequency	3.0 GHz
IF frequency	500 MHz
Mixer conversion loss	4.8 dB
RF BPF	3040–3520 MHz
IF BPF	450–550 MHz

Transmitter and Propagation Model :

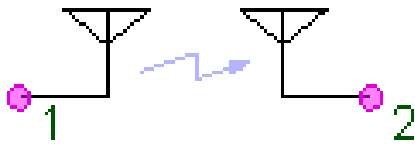
The RF source was implemented using the **MultiSource** component in SystemVue.

RF source parameters



MultiSource_1			
	ZFreq	0	Hz
S	PORT	1	
	R	50	Ω
	Name	Source1	
	Descrip		
	AutoDesc	1	
	Enable	1	
	Show	0	
	SrcType	0;0	
	Freq	3500	MHz
	Pwr	46	dBm
	Phase	0	°
	BW	0	MHz

The propagation path was implemented using the **AntPath model**.



AntPath_1			
S	G1	17	dB10
S	G2	0	dB10
	Lossa	20	dB10
	Lossb	103.32	dB10
	DIST	1	
	Loss1	2	dB10
	Loss2	5	dB10
	Zref	50.0	Ω
	ZIN	50.0	Ω
	ZOUT	50.0	Ω
	Ta	25	°C

The total path loss is:

$$\begin{aligned}
 L_{total} &= L_{FSPL} + G_{TX} + G_{RX} + L_1 + L_2 \quad \text{\#Link Budget Equation} \\
 &= 103.32 - 17 - 0 + 2 + 5 \\
 L_{total} &= 93.32 \text{ db}
 \end{aligned}$$

Therefore, the theoretical received power before the receiver front end is approximately:

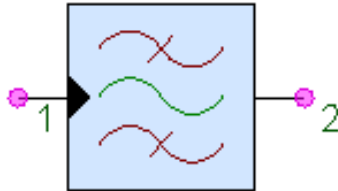
$$\begin{aligned}
 P_{RX} &= 46 - 93.32 \\
 P_{RX} &\approx -47.32 \text{ dbm}
 \end{aligned}$$

The actual SystemVue **DCP** (Delivered Carrier Power) result at the AntPath node was approximately: **-47.32 dbm**.

The difference occurs because SystemVue is evaluating the actual model response rather than simply applying the nominal hand calculation.

RF Bandpass Filter:

A third-order Butterworth bandpass filter was used as a representative RF preselector.



BPF_Butter_1			
S	IL	3.4	dB
S	N	3	
S	Flo	3040	MHz
S	Fhi	3520	MHz
	Apass	3.01	dB
	Amax	100	dB
	Zref	50	Ω
	Z1	50	Ω
	Z2	50	Ω
	TYPE	1	
	Ta	25	$^{\circ}\text{C}$

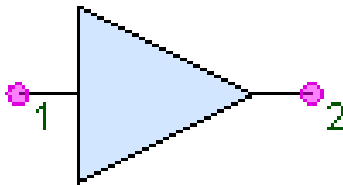
The purpose of the RF filter is to:

- Pass the desired 3.5 GHz RF signal
- Reject unwanted frequencies
- Reduce out-of-band interference before amplification and mixing.

Low-Noise Amplifier:

The LNA is placed immediately after the RF filter.

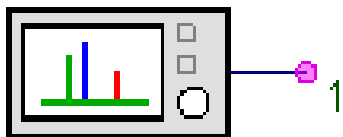
A representative LNA based on the characteristics of the **Analog Devices HMC392** was used.



RFamp_1			
S	G	15.5	dB
S	NF	2.4	dB
	OP1dB	16	dBm
	OPSAT	18	dBm
	OIP3	28	dBm
	OIP2	80	dBm
	RISO	50	dB
	MaxOrder	0:Use System Analysis setting	
	EnablePN	0:NO	
	NonLinearModelingMode	0:NonLinear Parameters	
	AMtoPM_Mode	0:Off	
	Zref	50	Ω
	PortParamType	0:Zin, Zout	
	ZIN	50	Ω
	ZOUT	50	Ω
	ISupply	0	mA
	VSupply	0	V
	FrequencyMode	0:Normal	
	ParamFreqList	[100]	MHz
	Ta	25	$^{\circ}\text{C}$

Local Oscillator:

A second **MultiSource** component was used as the local oscillator.

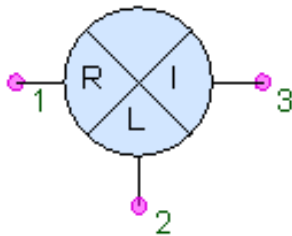


LO_Source			
	ZFreq	0	Hz
S	PORT	2	
	R	50	Ω
	Name	Source1	
	Descrip		
	AutoDesc	1	
	Enable	1	
	Show	0	
	SrcType	0;0	
	Freq	3000	MHz
	Pwr	7	dBm
	Phase	0	°
	BW	0	MHz

The LO is connected to the LO port of the mixer.

Mixer:

The mixer performs frequency conversion from RF to IF.



Mixer_1			
S	ConvGain	-4.8	dB
	SUM	0:Difference	
S	LO	7	dBm
	NF	5.3	dB
	InPort	0:RF	
	IP1dB	1	dBm
	IPSAT	2	dBm
	IIP3	11	dBm
	IIP2	21	dBm
	LOHarms	[30,10]	dB
	Show_More_Parameters	0:NO	
	FrequencyMode	0:Normal	
	ParamFreqList	[100]	MHz
	Show_Advanced_Parameters	0:NO	

The conversion gain is entered as **-4.8 dB** because the physical mixer has a **4.8 dB conversion loss**.

Mixer noise figure

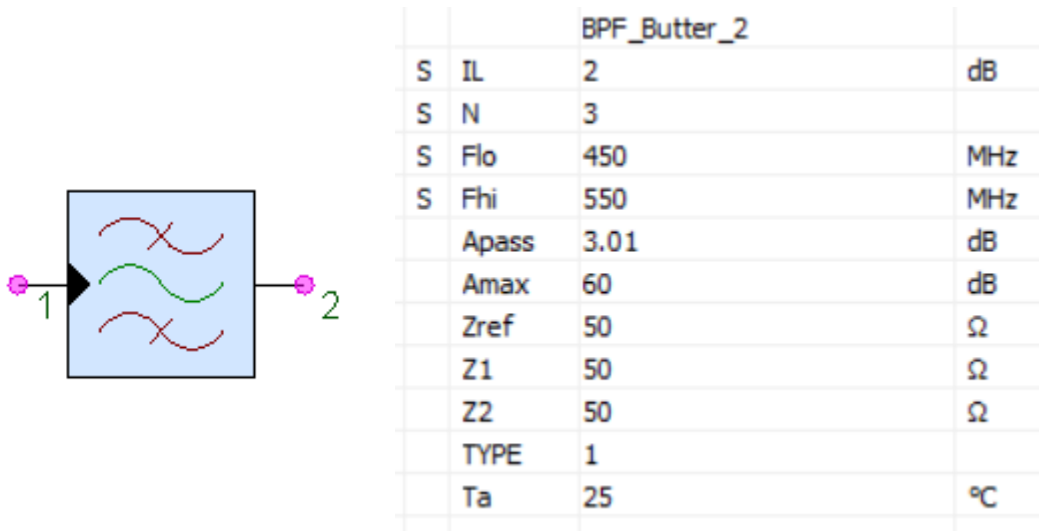
A representative value of: **NFmixer=5.3 dB** was subsequently used based on the typical relationship between passive mixer conversion loss and noise figure.

This should be explicitly described in the report as an **engineering estimate**, rather than a directly measured value.

IF Bandpass Filter:

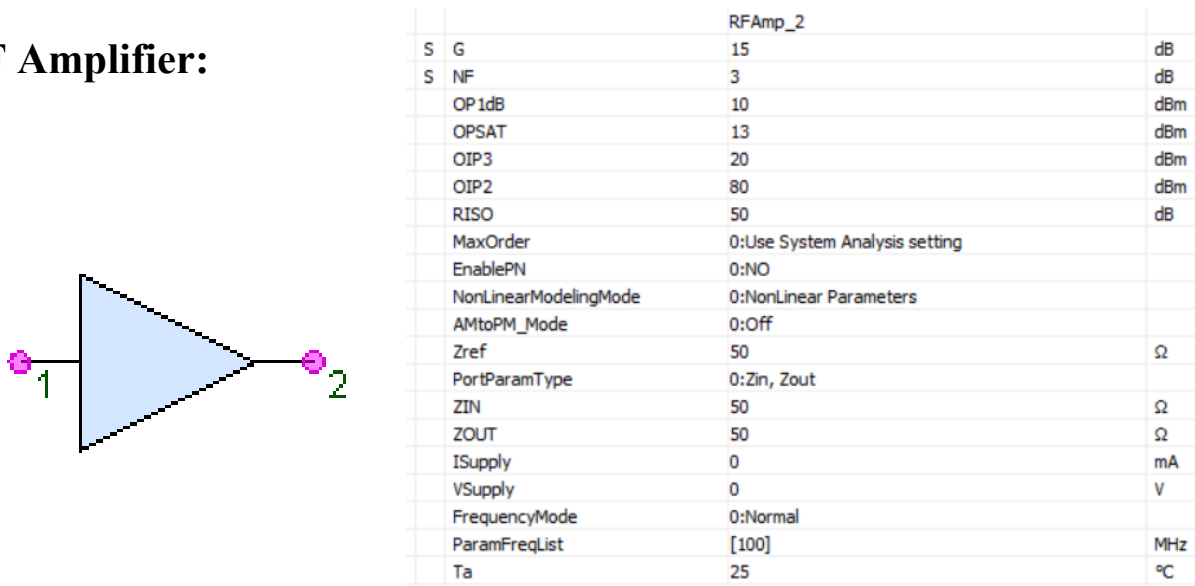
After mixing, the desired signal is at 500 MHz.

A representative third-order Butterworth IF filter was therefore implemented.



The filter passes the 500 MHz IF signal while rejecting unwanted mixer products.

IF Amplifier:



The purpose of the IF amplifier is to provide additional gain at the intermediate frequency before the receiver output.

RF System Analysis Path:

System Simulation Parameters

General

Paths

Calculate

Composite Spectrum

Options

Output

Design To Simulate:

Design1

Dataset:

☐ Automatic Recalculation

Frequency Units:

MHz

Measurement Bandwidth

Nominal Impedance:

50

Ohms

Channel:

1

MHz

Calculate Now

Save as Favorite...

Schematic Source Summary:

Name	Net Name	Description	
LO_Source	2	Universal Signal Source - Source1 =	Edit...
MultiSource_1	1	Universal Signal Source - Source1 =	Edit...

☐ Minimum number of source data points:

2

Factory Defaults

OK

Cancel

Apply

Help

System Simulation Parameters

General

Paths

Calculate

Composite Spectrum

Options

Output

Add All Paths From All Sources

Add Path

Delete All Paths

Name	Description	Enable		
Path1	Parts: 'MultiSource_1,Port_3' Frequency: 3500 MHz	☒	Edit...	Delete
			Add	

Factory Defaults

OK

Cancel

Apply

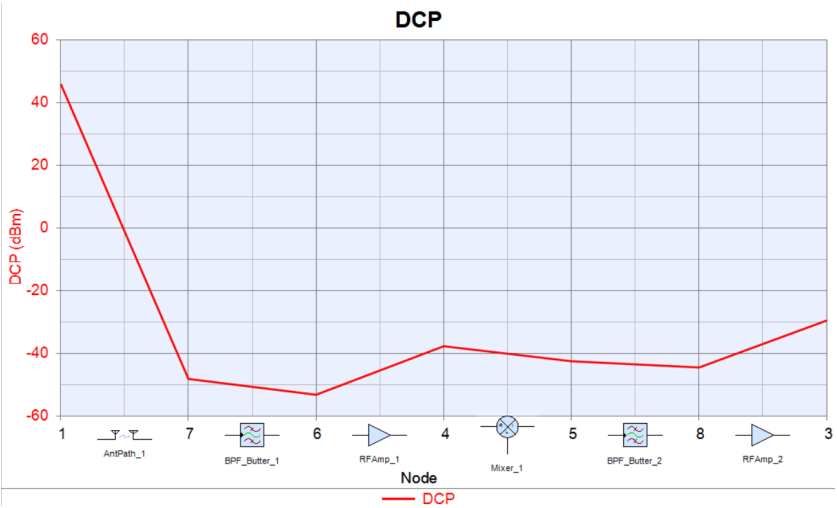
Help

SystemVue Power Results:

DCP

Delivered Carrier Power

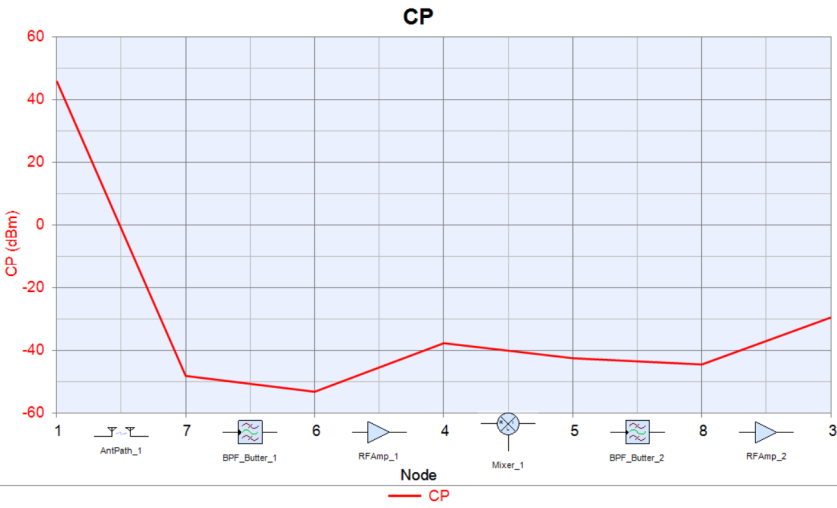
Index	deNames	DCP (dBm)
1	1	46
2	7	-48.1699
3	6	-53.219
4	4	-37.7189
5	5	-42.5191
6	8	-44.5114
7	3	-29.5114



CP

Carrier Power

Index	NodeNames	CP (dBm)
1	1	46.0043
2	7	-48.1656
3	6	-53.2146
4	4	-37.7146
5	5	-42.5147
6	8	-44.507
7	3	-29.5071



CGAIN

Carrier Gain

Index	NodeNames	CGAIN (dB)
1	1	0
2	7	-94.1699
3	6	-99.219
4	4	-83.7189
5	5	-88.5191
6	8	-90.5114
7	3	-75.5114

